

~~Choosing Better~~ Evaluating NLDL Layouts ~~by Evaluating the~~ using a Model in the High-dimensional Data Space

Jayani P. Gamage

Econometrics & Business Statistics, Monash University

and

Dianne Cook

Econometrics & Business Statistics, Monash University

and

Paul Harrison

MGBP, BDInstitute, Monash University

and

Michael Lydeamore

Econometrics & Business Statistics, Monash University

and

Thiyanga S. Talagala

Statistics, University of Sri Jayewardenepura

September 21, 2026

Abstract

Nonlinear dimension reduction (NLDL) techniques such as ~~tSNE~~, t-SNE and UMAP provide ~~a~~ low-dimensional representations of high-dimensional data ~~() by applying a nonlinear transformation. NLDL often exaggerates random patterns. But NLDL views have~~, capturing complex structures in a concise visual summary. While NLDL has an important role in data analysis ~~because, if done well, they provide a concise visual (and conceptual) summary of distributions. The NLDL methods and hyper-parameter choices can create~~, these methods can badly miss the mark by exaggerating structures or suggesting spurious patterns. Different methods hyper-parameters choices can produce wildly different representations, making it difficult to decide which ~~is best, or whether any or all are accurate or are layouts are reasonable representations and which are~~ misleading. To help ~~assess the NLDL and decide on which, if any, is the most reasonable representation of the structure(s) present in the data,~~ we have developed ~~an algorithm to show the NLDL model~~

~~in the space, viewed with a tour, a movie of linear projections. From this, one can see if the model fits everywhere, or better in some~~ a procedure that treats the low-dimensional NLDR layout as a model that can be lifted up into the original high-dimensional space. This enables us to assess goodness of fit by computing residuals and visualizing the fitted model overlaid on the data using grand tours. This approach allow us to quantitatively compare multiple layouts, determine which representations are reasonable, and assess whether an embedding captures the data structure throughout the space, only in certain subspaces, or ~~completely mismatches the data. Also, we can see how different methods may have similar summaries or quirks.~~ performs poorly overall.

Keywords: high-dimensional data, dimension reduction, data visualization, statistical graphics, tour, unsupervised learning

1 Introduction

Nonlinear dimension reduction (NLDR) is popular for making a convenient low-dimensional (k - D) representation of high-dimensional (p - D) data ($k < p$). Recently developed methods include t-distributed stochastic neighbor embedding (tSNE) (Maaten & Hinton 2008), uniform manifold approximation and projection (UMAP) (McInnes et al. 2018), potential of heat-diffusion for affinity-based trajectory embedding (PHATE) algorithm (Moon et al. 2019), large-scale dimensionality reduction using triplets (TriMAP) (Amid & Warmuth 2022), and pairwise controlled manifold approximation (PaCMAP) (Wang et al. 2021).

However, the representation generated can vary dramatically from method to method, choice of hyper-parameter, or even random seed, as illustrated by Figure 1. The specific method and hyper-parameters used to produce each layout (see Supplementary Materials) is not essential for the discussion. The dilemma for the analyst is which representation to use. The choice might choice may result in different procedures being used in the downstream analysis, or different inferential conclusions. Various academics have expressed concerns with about current practices and procedures for choosing (e.g., Irizarry (2024), Chari & Pachter (2023)). The research described here provides new numerical and visual tools to aid with this decision help choose a suitable layout.

The paper is organized as follows. Section 2 provides a summary of the literature on NLDR, gives more details about NLDR and high-dimensional data visualization methods. Section ?? contains the details of the. Sections 3, 4, 5 describe the new methodology, including illustrated using a simulated data example. In Section 6, we describe how to assess the best fit and identify the most accurate layout based on the proposed model.

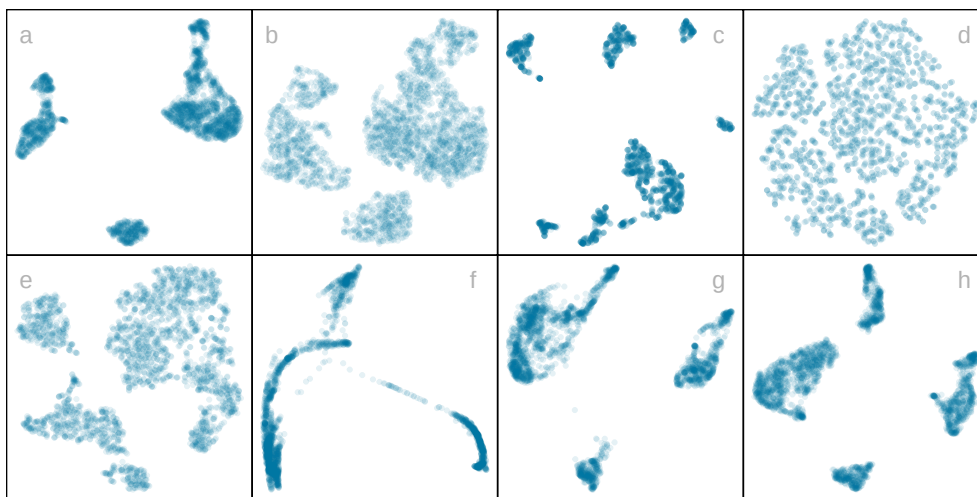


Figure 1: Eight different NLDR representations of the same data, produced by tSNE, UMAP, PHATE, TriMAP and PaCMAP with a variety of hyper-parameter choices. The variety in layouts makes it difficult to choose which best represents the data distribution.

~~diagnostics~~ explains how to use this approach to choose from a selection of layouts. [Section 7](#) extends the approach to predict the layout positions of new observations. Two applications illustrating the use of the new methodology for bioinformatics and image classification are in [Section 8](#). ~~Limitations and future directions are provided in~~ [Section 9](#) ~~discusses limitations and future directions~~. In addition, there is an R package [quollr](#) which implements the [methods](#).

2 Background [literature on dimension reduction](#)

~~Historically,~~ [Making useful](#) low-dimensional (k - D) ~~representations~~ [layouts](#) of high-dimensional (p - D) data ~~have been computed~~ [has historically been achieved](#) using multidimensional scaling (MDS) ([Kruskal 1964](#)); ~~which includes principal components analysis (PCA)~~ (for an overview see [Jolliffe \(2011\)](#)). (A contemporary comprehensive guide to MDS can be found in [Borg & Groenen \(2005\)](#).) ~~The representation can be considered to be a layout of points in~~ produced by an embedding procedure that maps the data from

. In MDS, the k -D layout is constructed by minimizing a stress function that ~~differences~~ compares distances between points in p -D with potential distances between points in k -D. Various formulations of the stress function result in non-metric scaling (Saeed et al. 2018) and isomap (Silva & Tenenbaum 2002). ~~Challenges in working with high-dimensional data, including visualization, are outlined in Johnstone & Titterton (2009)~~ Principal components analysis (PCA) produces the same layout as classical MDS if interpoint distances are Euclidean. Recent overviews of MDS, PCA and statistically conducting linear and nonlinear dimension reduction can be found in Borg & Groenen (2005), Johnstone & Titterton (2009), Jolliffe (2011), Koch (2013).

~~Many new methods for NLDR have emerged in recent years, all designed to better capture specific structures potentially existing in . Here we focus on five currently popular techniques:~~ The relatively recent NLDR methods, tSNE, UMAP, PHATE, TriMAP and PaCMAP, and PaCMAP, can be considered new approaches tackling the same problem as MDS. The methods tSNE, UMAP, TriMAP ~~and PaCMAP can be considered for producing~~ , and PaCMAP are similar in that they determine the k -D representation by minimizing the divergence between two inter-point distance distributions. PHATE is ~~an example of~~ different and is based on a diffusion process spreading to capture geometric shapes ~~, that~~ include both global and local structure. (See Coifman et al. (2005) for ~~an explanation of~~ more details on diffusion processes.)

The array of layouts in Figure 1 illustrate what can ~~emerge from the choices of method and~~ result from choosing different methods, tuning with different hyper-parameters, and even the random seed that initiates the computation. ~~Key structures interpreted from these views suggest~~ Each layout provides a different lens on the interpoint distances in the data.

The key structures in the high-dimensional data that we might hypothesize from these 2- D layouts are: (1) highly ~~separated clusters~~ *separated clusters* (a, b, e, g, h) with the number of clusters ranging from 3-6; (2) ~~stringy branches~~ *stringy branches* (f), and (3) ~~barely separated clusters~~ *barely separated clusters* (c, d) ~~which would contradict the other representations. These contradictions arise because these methods and hyper-parameter choices provide different lenses on the interpoint distances in the~~. They are at odds with each other, and it begs us to decide which are actually present in the p - D data.

The alternative approach to visualizing the high-dimensional data is to use linear projections. ~~PCA is the classical approach, resulting in a set of new variables which are linear combinations of the original variables, such as PCA.~~ Tours, defined by Asimov (1985), broaden the scope by providing movies of linear projections ~~, that provide views showing~~ *views of* the data from all directions. ~~(See Lee et al. (2021) for~~ *Lee et al. (2021) has* a review of tour methods. ~~) There are many tour algorithms implemented, with many~~ *Many tour algorithms are* available in the R package `tourr` (Wickham et al. 2011), and versions enabling better interactivity in `langevitour` (Harrison 2023) and `detourr` (Hart & Wang 2022). Linear projections are a safe way to view high-dimensional data ~~, because they do not warp the space, so they are more faithful representations of the structure. However, linear projections can be cluttered, and global patterns can obscure local structure. The simple activity of projecting data from p - D suffers from piling (Laa et al. 2022), where data concentrates in the center of projections. NLDR is designed to escape these issues, to exaggerate structure so that it can be observed. But as a result,~~ *NLDR can hallucinate wildly, to suggest patterns that are not actually present in the data.*

~~Our proposed solution is to~~ *Lee et al. (2022) provided interactive graphics to link between*

the NLDR view and tours, to help understand how the NLDR transformed the interpoint distances into a p -D layout. Here, we further develop this approach to directly use the tour to examine how the NLDR is warping the space. It follows what Wickham et al. (2015) describes as ~~model-in-the-data-space~~ model-in-the-data-space. The fitted model should be overlaid on the data, to examine the fit relative to the spread of the observations. While this is straightforward, and commonly done when data is 2-D, it is also possible in p -D, for many models, when a tour is used.

Wickham et al. (2015) provides several examples of models overlaid on the data in p -D. In hierarchical clustering, a representation of the ~~dendrogram~~ dendrogram using points and lines can be constructed by augmenting the data with points marking the merging of clusters. Showing the movie of linear projections ~~reveals~~ shows how the algorithm sequentially fitted the cluster model to the data. For linear discriminant analysis or model-based clustering, the model can be indicated by $(p - 1)$ -D ellipses. It is possible to see whether the elliptical shapes appropriately match ~~es~~ the variance of the relevant clusters, and to compare and contrast different fits. For PCA, one can display the model (a k -D plane of the reduced dimension) using wireframes of transformed cubes. Using a wireframe is the approach we take here, to represent the NLDR model in p -D.

3 ~~Method~~ Considering the NLDR layout as a model

3.1 ~~What is the NLDR model?~~

At first glance, thinking of NLDR as a modeling technique might seem strange. It is a simplified representation or abstraction of a system, process, or phenomenon in the real

world. The p -D observations are the realization of the phenomenon, and the k -D NLDR layout is the simplified representation. Typically, $k = 2$ ~~; which is used for the rest of this paper is used~~. From a statistical perspective, we can consider the distances between points in the 2-D layout to be the variance that the model explains, and the (relative) difference with their distances in p -D ~~is as~~ the error, or unexplained variance. We can also imagine that the positioning of points in 2-D represents the fitted values, ~~that which~~ will have some prescribed position in p -D that can be compared with their observed values. This is the conceptual framework underlying the more formal versions of factor analysis (Jöreskog 1969) and MDS. (Note that, for this thinking, the full p -D data needs to be available, not just the interpoint distances.)

We define ~~the NLDR as a function $g: \mathbb{R}^{n \times p} \rightarrow \mathbb{R}^{n \times 2}$, with an NLDR method as taking as input an $n \times p$ matrix $\mathbf{X}$, as well as user-specified hyper-parameters $\boldsymbol{\theta}$. These parameters, $\boldsymbol{\theta}$, depend on the choice of g , and can be considered part of model fitting in the traditional sense. Common choices for~~, and returning an $n \times 2$ matrix $\mathbf{Y}$. We will refer to rows i of $\mathbf{X}$ as $\mathbf{x}_{i*}$ and columns j as $\mathbf{x}_{*j}$, and similarly for other matrices. Conceptually, we will regard an NLDR method as first “fitting” a function $g: \mathbb{R}^p \rightarrow \mathbb{R}^2$, and then calculating the rows of the output matrix using $\mathbf{y}_{i*} = g(\mathbf{x}_{i*})$. The implementation of an NLDR method may not actually divide its working into these two steps. Some but not all NLDR method implementations provide access to an estimated ~~g include functions used in tSNE, UMAP, PHATE, TriMAP, PaCMAP, or MDS, although in theory any function that does this mapping is suitable,~~ allowing supplementary points in p -D to be mapped to 2-D.

With our goal being to make a representation of this 2-D layout that can be lifted into high-dimensional space, the layout needs to be augmented to include neighbor information.

A simple approach would be to triangulate the points and add edges. A more stable approach is to first bin the data, reducing it from n to $m \leq n$ observations, and connect the bin centroids. We recommend using a hexagon grid because it better reflects the data distribution and has ~~less~~ fewer artifacts than a rectangular grid ([Carr et al. 1987](#)). This process serves to reduce some noisiness in the resulting surface shown in p - D . The steps in this process are shown in Figure 2, and documented below.

To illustrate the method and how to use it to choose ~~a reasonable layout, we use simulated~~ from a selection of layouts, we simulated 7- D data, which we call the “2NC7” data. ~~It~~ , that has two separated nonlinear clusters, one forming a 2- D curved shape ([cluster 1](#)), and the other a 3- D curved shape ([cluster 2](#)), each consisting of 1000 observations. The first four variables, $x_{*1}, x_{*2}, x_{*3}, x_{*4}$ hold this cluster structure, and the remaining three ~~are purely noise. We would consider $T = (X_1, X_2, X_3, X_4)$ to be the geometric structure~~ (true model) x_{*5}, x_{*6}, x_{*7} are low-variance noise. Because no error was simulated in the first four variables, we would consider $[x_{*1}, x_{*2}, x_{*3}, x_{*4}]$ to be the geometric structure (true model) that we hope to capture. ~~This data is sufficiently simple, with just two complexities (two separated curvilinear clusters, and two different implicit dimensions), to adequately explain the new method. The applications section contains two practical examples where NLDR has been used in published work. This data has both global and local structure. In practice, all variables might have some small error, and a structure to detect might be in any or all variables. The 2NC7 data is simple but complex enough to illustrate the new methodology. We would consider the~~ two separated clusters ~~would be considered to be global structure~~ as global structure, and the nonlinear ~~low-dimensional shapes could be considered to be local structure, one being and the other . An ideal~~

~~NLDR layout would reveal the~~ surfaces to be relatively *local structure*. Figure 2 uses a reasonable 2-D NLDR layout of 2NC7, produced by tSNE, which reveals the two clusters with moderate separation ~~, and flatten (global), and flattens~~ the curvilinear forms while (hopefully) preserving the proximity of points ~~.(local)~~. Note that, because cluster 2 is 3-D, it cannot be fully captured in a 2-D layout, and the flat rectangle is as good as can be expected.

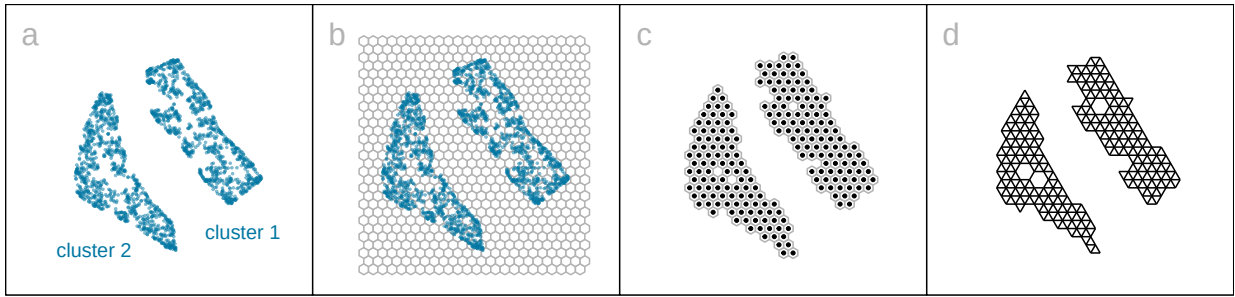


Figure 2: Key steps for constructing the model on the tSNE layout ($k = 2$) ~~of 2NC7~~: (a) data, (b) hexagon bins, (c) bin centroids, and (d) triangulated centroids. The layout used is a reasonable representation of the 2NC7 data~~is shown, which contains two curvilinear clusters that have been flattened into a triangular and rectangular shape.~~

3.1

4 Algorithm to represent the model in 2-D

4.0.1 ~~Scale the data~~

4.1 Scale the layout data

Because ~~we are working with the~~ algorithm is based on distances between points, ~~starting with data having a standard scale, e.g. scaling the layout data to range between~~ $[0, 1]$

~~in the first coordinate~~ is recommended. ~~The default should take~~ By default, the aspect ratio produced by the NLDR $(r_1, r_2, \dots, r_k)$ ~~into account~~ is preserved in the other coordinates. When $k = 2$, as in hexagon binning, the default range is $[0, y_{i,\max}]$, $i = 1, 2$, where $y_{1,\max} = 1$ and $y_{2,\max} = r_2/r_1$ (Figure 2). If the NLDR aspect ratio is ignored then set $y_{2,\max} = 1$. ~~for the second coordinate is $[0, r_2/r_1]$.~~

4.1.1 ~~Hexagon grid configuration~~

4.2 Hexagon grid configuration

Although there are several implementations ~~of hexagon binning~~ (Carr et al. 1987), and a published paper (Carr et al. 2023), surprisingly, none has sufficient detail or components that produce everything available (e.g., Carr et al. (2023)), the full set of data objects is needed for this project. So we ~~described the process used here.~~ have re-implemented the method so that all items needed for creating the p -D representation are available (illustrated in Figure 3 illustrates the notation used).

The 2- D hexagon grid is ~~defined by its bin centroids.~~ Each hexagon, H_h ($h = 1, \dots, b$) is uniquely described by centroid, $C_h^{(2)} = (c_{h1}, c_{h2})$. The number of bins in each direction is denoted as (b_1, b_2) , with $b = b_1 \times b_2$ being the total number of bins. We expect the user to ~~provide just~~ uniquely specified by the hexagon ids, $h (= 1, \dots, b)$, and their 2- D centroids, $c_{h*}^{(2)} = (c_{h1}, c_{h2})$. The algorithm inputs b_1 ~~and we calculate,~~ which determines b_2 using the NLDR ratio, ~~to compute the~~ producing $b = b_1 \times b_2$, the total number of bins in the grid.

~~To buffer is used to~~ ensure that the grid covers the range of data values ~~a buffer parameter~~, and provides flexibility for where the grid positining. It is well-known that where a grid

starts can change binning results, even though most documentation focuses on binwidth!

The buffer (q) is set as a proportion of the range. By default, $q = 0.1$.~~The buffer should be extending a full hexagon width (a_1) and height (a_2) beyond the data, in all directions, which is 10% of the range.~~ The lower left position where the grid starts is ~~defined as (s_1, s_2), and corresponds to set to be ($s_1 = -q, s_2 = -qr_2$), specifying the centroid of the lowest~~ lower left hexagon, $C_1^{(2)} = (c_{11}, c_{12})$.~~This must be smaller than the minimum data value. Because it is one buffer unit, q below the minimum data values, $s_1 = -q$ and $s_2 = -qr_2$.~~

$c_{1*}^{(2)} = (c_{11}, c_{12})$.

~~The value for b_2 is computed by fixing b_1 .~~ Considering the upper bound of the first NLDR component, ~~the horizontal binwidth must satisfy~~ $a_1 > (1 + 2q)/(b_1 - 1)$. Similarly, ~~for the second NLDR component, the vertical binwidth must satisfy~~

$$a_2 \geq \frac{r_2 + q(1 + r_2)}{(b_2 - 1)}.$$

$$a_2 \geq \frac{r_2 + q(1 + r_2)}{(b_2 - 1)}.$$
(1)

~~Since $a_2 = \sqrt{3}a_1/2$ for regular hexagons,~~ ~~Because we are using regular hexagons~~ $a_2 = \sqrt{3}a_1/2$, and Equation 1 can be re-written as

$$a_1 \geq \frac{2[r_2 + q(1 + r_2)]}{\sqrt{3}(b_2 - 1)} \frac{2(r_2 + q(1 + r_2))}{\sqrt{3}(b_2 - 1)}.$$

This is a linear optimization problem.~~Therefore, the optimal solution must occur on a vertex. Therefore,~~ ~~resulting in calculating b_2 as~~

$$b_2 = \left\lceil 1 + \frac{2[r_2 + q(1+r_2)](b_1-1)}{\sqrt{3}(1+2q)} \right\rceil.$$

$$\underline{b_2 = \left\lceil 1 + \frac{2(r_2 + q(1+r_2))(b_1-1)}{\sqrt{3}(1+2q)} \right\rceil.}$$

where $\lceil . \rceil$ indicates “smallest integer greater than or equal to”.

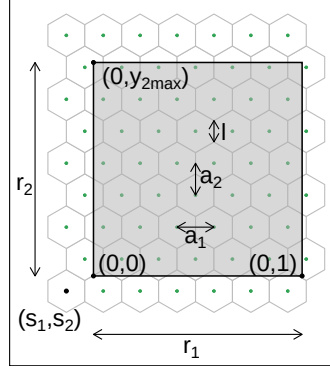


Figure 3: The components of the hexagon grid illustrating notation. The grey region indicates the extent of the data, which the grid must cover.

4.2.1 Binning the data

4.3 Binning the data

Observations are grouped into bins based on their nearest centroid. This produces a reduction in size of the data from n to m , where $\underline{m \leq b}$ (m (number of non-empty bins) $\leq b$ (total number of bins)). ~~This can be defined using the function $u: \mathbb{R}^{n \times 2} \rightarrow \mathbb{R}^{m \times 2}$, where~~ Each point i is assigned to a bin using

$$u(i)_i = \arg \min_{j=1, \dots, b} \sqrt{(y_{i1} - C_{j1}^{(2)})^2 + (y_{i2} - C_{j2}^{(2)})^2}, \underline{h=1, \dots, b} \sqrt{(y_{i1} - c_{h1}^{(2)})^2 + (y_{i2} - c_{h2}^{(2)})^2}$$

~~maps observation i into $H_h = \{i | u(i) = h\}$.~~

By default, the bin centroid is used for describing a hexagon (as done in Figure 2 (c)), but any measure of center, such as a mean or weighted mean of the points within each hexagon, could be used and then collected into sets $H_h = \{i | u_i = h\}$ ($h = 1, \dots, b$). Then $\hat{y}_{i*} = c_{u_i*}^{(2)}$ ($i = 1, \dots, n$) are considered the fitted values, from which we can also calculate 2- D residuals. The bin centers, and the binned data, are the two important components needed to render ID, u_i , for each observation is also needed to construct the model representation in high dimensions.

4.3.1 Indicating neighborhood

4.4 Indicating neighborhood

Delaunay triangulation (Lee & Schachter 1980, Gebhardt et al. 2024) is used to connect points so that edges centroids to indicate neighboring observations, ~~in both in~~ the NLDR layout (Figure 2 (d)) ~~and the model representation. When the data has been binned the triangulation connects centroids. The edges.~~ These are retained to preserve the neighborhood information ~~from the representation~~ when the model is lifted into p - D .

4.5

5 The model in p - D

~~The last~~

5.1 Lifting from 2- D to p - D

The next step is to lift the 2- D model into p - D by computing p - D vectors that represent bin centroids. We use the p - D mean of the points in a given hexagon, H_h , denoted $\mathbf{c}_h^{(p)}$, to map the centroid $\mathbf{c}_h^{(2)} = (c_{h1}, c_{h2})$ to a point in p - D . Let the j^{th} component of the p - D mean be

$$\mathbf{c}_{hj}^{(p)} = \frac{1}{n_h} \sum_{i=1}^{n_h} x_{hij}, \quad h = 1, \dots, b; \quad j = 1, \dots, p; \quad n_h > 0.$$

Then $\hat{\mathbf{x}}_{i*} = \mathbf{c}_{u_i}^{(p)}$ ($i = 1, \dots, n$) are considered the fitted values, from which we can also calculate p - D residuals. We could conceptualise this as a pseudo-inverse g^- , because for any 2- D observation, $\mathbf{y}_{i*}$ in a non-empty bin, we can provide coordinates for it in p - D , which would correspond to $\hat{\mathbf{x}}_{i*}$.

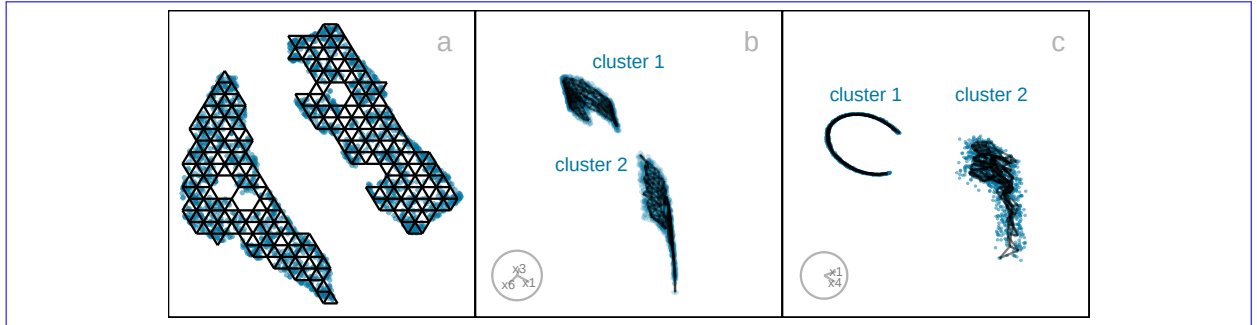


Figure 4: Lifting the 2- D fitted model (a) into p - D (b,c). Two projections of the p - D fitted model overlaying the data are shown in b, c. The fit is reasonably tight with the data in one cluster (top one in b), but slightly less so in the other cluster 2, probably because it is 3- D . Notice also that, in the 2- D layout, the two clusters have internal gaps which creates a model with some holes. This lacy pattern happens regardless of the hyper-parameter choice for tSNE, but this doesn't severely impact the p - D model representation.

5.2 Measuring the fit

All NLDR methods internally optimize a quantity an objective function to produce a layout for any particular hyper-parameter set. These a given set of hyper-parameters. These

objective functions are not always made available to the user in the model output ~~;~~ ~~and may~~ ~~not be universally comparable between hyper-parameter choices and~~ and or comparable across methods.

~~Several common metrics are often~~ A range of established metrics is commonly used to assess the quality of ~~any NLDR layout, based on preservation of global and local structure of the data. The NLDR embeddings, each emphasizing different aspects of structure preservation.~~ For example, the RNX curve ~~quantifies the~~ summarizes neighborhood agreement between p - D and k - D spaces ~~;~~ ~~by computing the~~ across a range of scales, with its area under the curve (~~$ARNX$~~) ~~across a range of neighborhood scales~~ (Lee et al. 2015). ~~A high value indicates better preservation of a balance of global and local structure~~ $ARNX$ reflecting a balance between local and global structure preservation (Lee et al. 2015). Random Triplet Accuracy (RTA) ~~and Centroid Triplet Accuracy compare the order of~~ evaluates the consistency of distance orderings among triplets of points in 2 - D and p - D ~~distances of random triplets of points~~ (Wang et al. 2021). ~~High values indicate preservation of the geometry, suggesting both local and global structure preservation.~~ spaces, providing a measure of geometric preservation (Wang et al. 2021). The Shepard diagram (Shepard 1962) ~~and,~~ together with its associated Spearman correlation (SC) (Spearman 1961) ~~between and distances.~~ assesses the monotonic relationship between pairwise distances in the two spaces and is often interpreted as a measure of global structure preservation. The Global Score (GS) ~~measures how well an embedding retains the overall geometry of the data relative~~ compares the embedding to a PCA baseline (Amid & Warmuth 2022). ~~Higher values indicate better preservation of global structure. The metric RTA, SC, GS, and $ARNX$ have been reversed~~ and evaluates how well the overall

geometry is retained (Amid & Warmuth 2022).

These metrics provide useful but complementary perspectives, as they capture different and often competing objectives, such as preserving local neighborhoods versus global distances. As a result, it is common for these measures to rank embeddings differently, reflecting the multi-objective nature of NLDR rather than a deficiency of any single metric.

To facilitate comparison with our proposed measure, we use reversed forms of these metrics (rRTA, rSC, rGS, and rARNX) ~~so that they align with HBE – the lower the value the better the layout,~~ so that lower values consistently indicate better performance.

~~None of~~ While the above measures ~~is particularly well-suited to assessing our model fit, as we will show later. Thus, we need a different approach to measuring model fit. Because the model here is similar to a confirmatory factor analysis model (see a general explanation in Brown (2015)), $\hat{T} + E$, our approach is similar to the ones used in this area. it is based on “residuals” computed as the difference between the fitted model and observed values~~ evaluate properties of the embedding itself, our goal is different: to assess how well the embedding supports a representation of the data in the original p - D space. This motivates a complementary approach based on model residuals.

We treat the embedding as defining a fitted representation in p - D . ~~Observations are associated with their bin center, $C_h^{(p)}$, which are also considered to be the fitted values. In factor analysis language, these fitted values might also be denoted as $\hat{X}$. The error is computed by taking the squared Euclidean distance, of points from their~~ Each observation $\mathbf{x}_{i*}$ has an associated bin centroid, which we will call hexbin error (HBE) u_i , with a centroid that serves as its fitted value, $\hat{\mathbf{x}}_{i*} = \mathbf{c}_{u_i*}^{(p)}$. The discrepancy between observed and fitted

values can then be interpreted as a residual.

We define the hexbin error (HBE) as the root mean squared Euclidean distance between observations and their associated bin centroids:

$$\text{HBE} = \sqrt{\frac{1}{n} \sum_{i=1}^n \sum_{j=1}^p (x_{ij} - \hat{x}_{ij})^2} \quad (2)$$

where n is the number of observations, m is the number of non-empty bins, n_h is the number of observations in h^{th} bin, p is the number of variables and $\mathbf{x}_{hij}$ is the j^{th} dimensional data of i^{th} observation in h^{th} hexagon. We can consider $e_{hi} = \sqrt{\sum_{j=1}^p (\mathbf{x}_{hij} - \mathbf{C}_{hj}^{(p)})^2}$ to be the residual for each observation. Figure 5 shows plots of $e_i = \sqrt{\sum_{j=1}^p (x_{ij} - \hat{x}_{ij})^2}$ as a density (a), coloring the points as colors in the NLDR layout (b) and the points, and in a tour (c). It can be seen that the biggest residuals are concentrated in one cluster, which occurs due to reflecting the intentional design that one cluster is slightly this cluster has an intrinsic higher-dimensional structure (approximately 3-D and perfectly captured by) and therefore cannot be fully represented in a 2-D layout.

Examining the distribution of residuals in a jittered dotplot (a), NLDR layout (b) and a tour of data space (c). Color indicates residual (e_{hi}), dark color indicating high value. Most large residuals are distributed in one cluster (bottom one in c) and most small residuals are distributed in the other cluster.

5.3

NLDR methods are primarily designed for visualization and exploration rather than reconstruction, and do not explicitly provide out-of-sample prediction. Of the five methods studied here, only UMAP provides a `predict()` function for embedding new data points based on the learned manifold (Konopka 2023). Several other approaches, not used here, PCA, neural network autoencoders (Hinton & Salakhutdinov 2006) and parametric tSNE (van der Maaten 2009) support prediction.

A benefit of our approach is that for any NLDR method embedding. Importantly, HBE is not intended to replace existing embedding-quality metrics. Instead, it provides a way to predict the layout position of a new observation, x' . The steps are (1) determine the closest bin centroid in $C_h^{(p)}$ and (2) predict the embedding to be the bin centroid in $C_h^{(2)}$. complementary diagnostic perspective, highlighting where the embedding fails to represent the original data structure in p -D. Differences between HBE and other metrics can therefore be interpreted as informative signals about trade-offs between local and global structure preservation.

5.3 Tuning

The model fitting is based on several parameters, including the hexagon bin parameters and the low count bin removal process. The hexagon bin parameters define the bottom-left bin position (s_1, s_2) , the number of bins in the horizontal direction (b_1) , which also determines the number of bins in the vertical direction (b_2) , the total number of bins (b) , and the total number of non-empty bins (m) . Low count bins are removed using standardized bin counts, defined as $w_h = n_h/n$, $h = 1, \dots, m$.

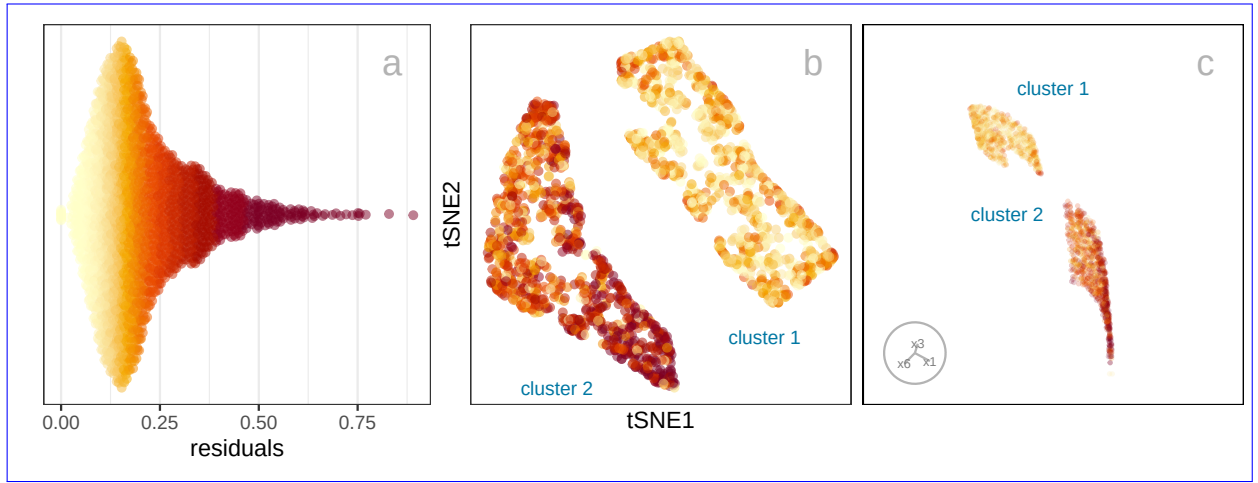
Default values are provided for each of these, but deciding on the best model fit is assisted by examining a range of values. The default number of bins $b = b_1 \times b_2$ is computed based on the sample size, by setting $b_1 = n^{1/3}$, consistent with the Diaconis-Freedman rule (Freedman & Diaconis 1981). The value of b_2 is determined analytically by b_1, q, r_2 (Equation ??). Values of b_1 between 2 and $b_1 = \sqrt{n/r_2}$ are recommended, where the dependence on r_2 reflects the preservation of aspect ratio in the NLDR layout.

Figure ?? shows the hexbin grids for three choices of b_1 . While the number of bins is the common parameter to modify, bin start positions (s_1, s_2) can be worth experimenting with also because it can also change bin counts.

5.3 ~~Interactive~~ Diagnosing the fit with interactive graphics

Matching points in the 2- D layout with their positions in p - D is useful when ~~tuning the fit~~. This can be used to examine the fitted model in some subspaces in , in particular in ~~association with residual plots~~.

assessing the fit, deciding on a layout, and basically understanding what a layout communicates about the data. The interactive 2- D layout (Sievert 2020) and the langevitour (Harrison 2023) view with the fitted model overlaid can be linked using a browsable HTML widget (Cheng & Sievert (2025), Cheng et al. (2024)). A rectangular “brush” is used to select points in one plot, which will highlight the corresponding points in the other plot(s). Because the langevitour is dynamic, brush events that become active will pause the animation, so that a user can interrogate the current view. This approach ~~will be illustrated on the examples~~ is illustrated in the examples in the Appendix and video supplements, to show how it can help to understand how ~~the an~~ NLDR has organized the



Hexbin density plots of tSNE layout of the 2NC7 data, using three different b_1 specifications yielding different b_2, b, m : (a) **15**, 18, 270, 98, (b) **24**, 29, 696, 209, and (c) **35**, 42, 1470, 386. Color indicates standardized counts, dark indicating high count and light indicates low count. At the smallest binwidth, the data structure is discontinuous, suggesting that there are too many bins.

It is worthwhile to consider what are desirable aspects of a hexbin result, that maps to summarizing the fit well. The binning should capture the underlying data distribution closely, with minimum number of necessary bins. An ideal binning might be indicated by a more uniform distribution of bin counts, or having few relatively empty bins. To help with this assessment average bin count ($\bar{n} = \sum n_h/m$), average standardized bin count ($\bar{w} = \sum w_h/m$) and proportion of non-empty bins (m/b), are also computed. Figure ?? shows some choices of plots of these quantities for a single NLDR layout, with three choices of a_1 indicated. Some expectations and reasoning for these plots is:-

- HBE will increase as a_1 increases, so good choices will be just before a big increase. In plot a, HBE changes fairly steadily so there is no easy choice to make.
- HBE can also be examined against average standardized bin count or average bin count (plot b). This is similar to the comparison with a_1 but to use when comparing different NLDR layouts. Different layouts might produce different density of points, which will not be captured well by a comparison of HBE vs a_1 .
- The proportion of non-empty bins is interesting to examine across different binwidths (plot c). A good binning should have just the right amount of bins to neatly cover the shape of the data, and no more or less. As binwidth gets smaller, m/b should roughly get bigger.
- Bins with a small number of observations might be removed to sharpen the wireframe model. This can have adverse effects, though - failing to extend the wireframe into sparse areas, or resulting in holes in the wireframe. Plot d shows the relationship between HBE (computed for all observations despite some bin removal) and the standardized bin count cutoff used to remove bins. For all three chosen bin widths a small number of bins can be removed without affecting HBE.

Figure 5: Various plots to help tune Examining the model fit: distribution of residuals in a jittered dotplot (a) HBE vs a_1 , 2-D NLDR layout (b) HBE vs $\bar{n}_h$, and a tour of 4-D data space (c) proportion of non-empty bins (m/b) vs a_1 , (d) HBE vs w_h cutoff used for removing low count bins. Color indicates the three binwidths show in Figure 6: 0.03 residual value (orange dashed e_i), 0.05 (blue solid), and 0.08 (green dotted) dark indicating high value. The better model fit will have low HBE Most large residuals are distributed in cluster 2, but reasonably sized bins that which is due to the 2-D's inability to capture the data sufficiently 3-D structure. Proportion of non-empty bins tends to increase with a_1 (c). Removing a few low count bins doesn't substantially change HBE, for all three binwidths

observations, and learn where it does not do well.

6 Choosing from a selection of layouts

Figure 6 illustrates the approach to compare the fits for different representations and assess the strength of any fit. What does it mean to be a best fit for this problem? Analysts use an NLDR layout to display the structure present in high-dimensional data in a convenient display. It is a competitor to linear dimension reduction that can better represent nonlinear associations such as clusters. However, these methods can hallucinate, suggesting patterns that don't exist, and grossly exaggerate other patterns. Having a layout that best fits the high-dimensional structure is desirable but more important is to identify bad representations so they can be avoided. The goal is to help users decide on a shows the diagnostic plot of HBE vs binwidth (i) for a selection of plots from which to use. Color in all plots indicates the layout (a-f). The details of how they were generated are in the Appendix. There is nothing special about this selection of layouts other than that all five methods were used: (a, f) tSNE, (b) UMAP, (c) PHATE, (d) TriMAP, (e) PaCMAP. They represent different patterns, but all agree on there being two clusters, with varying distance and shape between them. The process to make a decision is:

1. Fit the model to a range of binwidths for each layout, calculating HBE.
2. Plot each HBE against binwidth, for all layouts.
3. The better representation(s) will have universally lower HBE. When the lines cross, one needs to decide whether the local structure (smaller binwidth) is more important than the larger structure (large binwidth).

4. Examine the model for the chosen layout in p -D. The exact choice of binwidth isn't important, but we recommend a reasonably small one, so that the ~~most useful~~ and appropriate low-dimensional representation of the high-dimensional wireframe is smoother, and there is a chance to see small details of departures between the fitted model and the data. (See Figure 4 b, c for this data.)

~~A particular pattern that we commonly see is that analysts tend to pick layouts with clusters that have big separations between them. When you examine their data in a tour, it is almost always that we see there are no big separations, and actually often the suggested clusters are not even present. While we don't expect that analysts include animated gifs of tours in their papers, we should expect that any~~ For this set of layouts, either a or f would be preferable to the other layouts, because the HBE values are lower at every binwidth. (Note that a was used to illustrate the approach in the previous sections.)

Using binwidth as the horizontal axis is a choice. An alternative is to use the average bin count. In theory, this is the more sensible choice, because it summarizes the grid occupancy of the data. Using binwidth has the problem that a smaller binwidth for one layout might be equivalent to a larger binwidth in a different layout, as a fraction of the observations. Using binwidth sets up a possible mismatching HBE value. However, using average bin count tends to lose the ability to penalize layouts that exaggerate inter-cluster distance (see plots in the Appendix). An argument for why binwidth is the better choice is that it treats the ~~2-D representation adequately indicates the clustering that is present, and honestly show lack of separation or lack of clustering when it doesn't exist. It is important for analysts to have tools to select the accurate representation not the pretty but wrong representation.~~ space as a uniform canvas on which to compare the representations.

~~To decide on a reasonable layout, an analyst needs a selection of NLDR representations generated using a range of hyper-parameter choices and possibly different methods, such as tSNE and UMAP. They also require a range of model fits created by varying the binwidths and the level of low count bin removal, along with the calculated HBE values for each layout after transformation into the~~ It can also be useful to calculate the suite of other metrics to measure the fit, for each layout, which is done in plot ii. The comparison with HBE requires that a single value be chosen. Here, $a_1 = 0.05$ is used, but the actual choice wouldn't substantially matter because the ordering of the curves is quite consistent across binwidths. Note that in this parallel coordinate plot, the individual metrics are individually scaled to range between 0 and 1. HBE shows most agreement with rARNX, both rank a and f the best. They differ in how they rate (1) layout d - HBE would consider it the worst, and rARNX to be not quite as good as a and f - and (2) layout c - rARNX considers it to be the worst, and HBE in the middle. The other three metrics roughly agree with each other and are almost the polar opposite of HBE. This is most likely explained by the tension between local and global structure. Most will prefer a bigger distance between clusters in the layout, or at least not penalise this. The problem is that this leads to incorrect perception when the clusters are not widely separated, as is the case for this data (Figure 4).

The final check is to examine the chosen model fit (or several top-rated fits) in the p - D space. Finally, the analyst must be able to visually examine how well each model fits the data in the original data space. ~~data space~~ Figure 4 (b, c). As with all model fitting, if the data closely hugs the model representation, in all projections shown by the tour, then the choice is further supported. This is a suitable approach to better understand why metrics disagree on layouts. The Appendix has a comparison of the model fits for layouts

a and c. Layout c twists extremely along the 2- and 3- D manifolds where the data lies. Twisting in the fitted model appears to be very common for all methods for many data sets, which one could imagine as a solution for squashing higher dimensions into lower ones while preserving the neighborhood. In this case, it is quite extreme.

Comparing the HBE to obtain the best fit is appropriate if the same NLDR method is used. However, because the HBE is computed on You might find that some elements of the p - D data it measures the fit between model and data so it can also be used to compare the fit of different NLDR methods. A lower HBE indicates a better NLDR representation. structure are captured, but others are not. If this happens, a strategy is to break the data into subsets and repeat the process of deciding on a layout. Showing multiple layouts for a data set may better illustrate its nature. This might be advisable with 2NC7 also. At first glance, layout b looks preferable, because the distribution of observations within each cluster is smoother than in either (a) or (f). This likely reflects that the layout better captures the local structure.

Figure 6 compares the metrics rARNX, rRTA, rSC, rGS, along with HBE computed on $\alpha_1 = 0.05$ for the six layouts shown in Figure 6. This is a parallel coordinate plot where the y-axis shows a normalized score to ensure the metrics are on the same scale. Each line corresponds to one layout.

7 Prediction into 2- D

There is some agreement between the metrics. All, except rARNX and HBE agree that layout d is best. rARNX and HBE agree that layout f is best or very close to best. Layout

a is best according to HBE and rARNX but considered to be much less optimal by rRTA, rSC and rGS. Layout e is considered poor by rRTA, rSC, rGS, and rARNX. This illustrates how difficult it is to use the numerical metrics alone to decide on the best layout.

The problem with rSC is that correlation is not a good measure in the presence of clusters – the further the clusters are apart in the layout produces a Shepard plot with two clusters of distances which will produce a high correlation value. Similar reasoning would explain why rRTA and rGS behave similarly: they put too much emphasis on the global structure. Thus, for the 2NC7 data further apart clusters score better, overly emphasizing that there are two clusters, even though this separation is not accurately reflecting the difference in.

Only UMAP provides a way to predict the layout in-to-compare-with positions of new observations (Konopka 2023) when the full data is available, and this is made available in the umap (McInnes et al. 2018) software. Other approaches for embedding new data include neural network autoencoders (Hinton & Salakhutdinov 2006) and parametric tSNE (van der Maaten 2009) but they do not directly relate to the common methods presented here.

A benefit of our approach is that with the observed data, it can help to add more clarity in making decisions. Figure ?? shows the fitted models for layouts a model representation, we can predict layout positions for any method. To illustrate this, we will first compare the approach with the predictions provided by UMAP, by generating a test set of 2NC7. We will also show predictions of this test set for the layouts used in Figure 6.

Our procedure for the predicting the layout position of a new observation is: (i) identify

the closest bin centroid in p -D, $\mathbf{c}_{h*}^{(p)}$, and (rated high by HBE and rARNX) and $\mathbf{c}$ (rated poorly). These are ii) assign the corresponding centroid in 2-D projections from $\mathbf{c}_{h*}^{(2)}$, as the predicted embedding. This is not entirely different from how UMAP provides predictions, which is to find the nearest neighbors in p -D, and interpolate their positions in 2-D to provide the predicted embedding. This can be considered to be an estimate of the mapping g that maps the new observation ($\mathbf{x}^*$) into the layout ($\mathbf{y}^*$), $\mathbf{y}^* = \hat{g}(\mathbf{x}^*)$, $\hat{g}: \mathbb{R}^p \rightarrow \mathbb{R}^2$.

Figure 7 shows the results for UMAP (a) and our predictions (b). The 2NC7 data is divided into training and test sets, using a 70% split. The training data is used to make the 2-D layout. (This is similar but not quite the same as the layout on the full data seen in Figure 6 plot b.) The test set is held out for prediction. In both cases, it can be seen that the tour, with black indicating the fitted model overlaid on the blue points of the data. The reason for the poor fit is that the PHATE layout (c) twists extremely along the 2-manifolds where the data lies. We have learned that all the NLDR methods tend to have twists in the fit in but this is extreme. This is likely why layout c has poor metrics relative to the other layouts, and it suggests that it does not adequately capture the local structure in the 2NC7 data. The test set matches the training set in the layout quite well. The main differences are that our predictions are to the bin centroid, so it looks gridded. Adding some jitter to the predicted values would make them look more similar to the UMAP predictions. Just because we can, doesn't mean we should - for this data, the method provides demonstrably reasonable results, and better than having no method at all.

Three 2-D projections from a tour showing the fitted models (black lines) for layouts a (top row) and c (bottom row) of the 2NC7 data (blue points). Layout c, which was poorly rated by rARNX and HBE covers less of the width of the data than a. The triangular gridding

is less visible in c than layout a, and actually corresponds to extreme twisting.

8 Applications

To illustrate the approach, we use two examples: PBMC3k data (single cell gene expression), where an NLDR layout is used to represent cluster structure present in the p - D data, and MNIST hand-written digits, where NLDR is used to represent ~~essentially~~ a low-dimensional nonlinear manifold in p - D .

8.1 PBMC3k

This is a benchmark single-cell RNA-Seq data set collected on Human Peripheral Blood Mononuclear Cells (PBMC3k) as used in [10x Genomics \(2016\)](#). Single-cell data measures the gene expression of individual cells in a sample of tissue (see, for example, [Haque et al. \(2017\)](#)). This type of data is used to obtain an understanding of ~~cellular-level~~ cellular-level behavior and heterogeneity in their activity. Clustering of single-cell data is used to identify groups of cells with similar expression profiles. NLDR is often used to summarize the cluster structure. Usually, NLDR does not use the cluster labels to compute the layout, but uses color to represent the cluster labels when it is plotted.

In this data, there are 2622 single cells and 1000 gene expressions (variables). Following the same pre-processing as [Chen et al. \(2024\)](#), different NLDR techniques were performed on the first nine principal components. Figure 1 shows this data using a variety of methods, and different hyper-parameters. You can see that the result is wildly different depending on the choices. Layout a is a reproduction of the layout that was published in [Chen et al.](#)

(2024). This layout suggests that the data has three very ~~well-separated~~ well-separated clusters, each with an odd shape. The question is whether this accurately represents the cluster structure in the data, or whether they should have chosen ~~bor-eor-dor-eor-for-g~~, c, d, e, f, g, or h. This is what our new method can help with – to decide which is the more accurate $2-D$ representation of the cluster structure in the $p-D$ data.

Figure 8 shows HBE across a range of binwidths (a_1) for each of the layouts in Figure 1. The layouts were generated using tSNE and UMAP with various hyper-parameter settings, while PHATE, PaCMAP, and TriMAP were applied using their default settings. Lines are ~~color-coded~~ color-coded to match the color of the layouts shown on the right. Lower HBE indicates ~~the a~~ a better fit. Using a range of binwidths shows how the model changes, with possibly the best model being one that is universally low HBE across all binwidths. It can be seen that layout f is sub-optimal with universally higher HBE. Layout a, the published one, is better, but it is not as good as layouts b, d, or e. With some imagination, the layout d perhaps shows three barely distinguishable clusters. Layout e shows three, possibly four, clusters that are more separated. The choice reduces from eight to these two. Layout d has slightly better HBE when the a_1 is small, but layout e beats it at larger values. Thus, we could argue that layout e is the most accurate representation of the cluster structure ~~;~~ of these eight.

To further assess the choices, we need to look at the model in the data space, by using a tour to show the wireframe model overlaid on the data in the $9-D$ space (Figure 9). Here we compare the published layout (a) versus what we argue is the best layout (e). The top row (a1, a2, a3) corresponds to the published layout, and the bottom row (e1, e2, e3) corresponds to the optimal choice according to our procedure. The middle and right plots

show two projections. The primary difference between the two models is that the model of layout e does not fill out to the extent of the data but concentrates in the center of each point cloud. Both suggest that three clusters ~~is~~ are a reasonable interpretation of the structure, but the layout e more accurately reflects the separation between them, which is small.

~~Assessing which of the 8 NLDR layouts on the PBM3k data (shown in Figure 1) is the better representation using HBE for varying (i) binwidth (a_1), and (ii) average bin count ($\bar{n}_h$). Color used for the lines and points in the left plot and in the scatterplots represents NLDR layout (a-h). Layout f is universally poor. Layouts a, c, g, h that show large separations between clusters are universally suboptimal. Layout d with little separation performs well at tiny binwidth (where most points are in their own bin) and poorly as binwidth increases. The choice of best is between layouts b and e, that have small separations between oddly shaped clusters. Layout e is chosen as the best. Plot (ii), which accounts for the density within clusters by using average bin count, shows reduced differences between layouts, indicating that part of the variation in (i) is driven by cluster density rather than true structural differences.~~

~~Compare the published layout (a) made with UMAP and the layout selected by HBE plot (e) made by tSNE. The two plots on the right show projections from a tour, with the models overlaid. The published layout a suggested three very separated clusters, but this is not present in the data. While there may be three clusters they are not well-separated. The difference in model fit also indicates this: the published layout a does not spread out fully into the point cloud like the model generated from layout e. This supports the choice that layout e is the better representation of the data, because it does not exaggerate separation~~

~~between clusters.~~

8.2 MNIST hand-written digits

The digit “1” of the MNIST dataset (LeCun et al. 1998) consists of 7877 grayscale images of handwritten “1”s. Each image is 28×28 pixels, which corresponds to 784 variables. The first 10 principal components, explaining 83% of the total variation, are used. This data essentially lies on a nonlinear manifold in ~~the~~-high dimensions, defined by the shapes that “1”s make when sketched. We expect that the best layout captures this type of structure and does not exhibit distinct clusters.

Figure 10 compares the fit of six layouts computed using UMAP (b), PHATE (c), TriMAP (d), PaCMAP (e), with default hyper-parameter setting and two tSNE runs, one with default hyper-parameter setting (a) and the other changing perplexity to 89 (f). The layouts are reasonably similar in that they all have the observations in a single blob. Some (b, c) have a more curved shape than others. Layout e is the most different, having a linear shape ~~and~~ and a single very large outlier. Both a and f have a small clump of points, perhaps slightly disconnected from the other points, in the lower to middle right.

The layout plots are colored to match the lines in the HBE vs binwidth (a_1) plot. Layouts a, b, and f fit the data better than c, d, e, and layout f appears to be the best fit. Figure 11 shows this model in the data space in two projections from a tour. The data is curved in the 10- D space, and the fitted model captures this curve. The small clump of points in the 2- D layout is highlighted in both displays. These are almost all inside the curve of the bulk of points and are sparsely located. The fact that they are packed together in the 2- D layout is likely due to the handling of density differences by the NLDR.

An interesting aside is that the rather strange layout e, which has what looks like a single point far from the remaining observations, is actually similar to this one. That point is actually a clump of points corresponding to some of the diffuse points interior to the curve of the bulk of points. This is easy to see using the linked brushing tool.

The next step is to investigate the 2- D layout to understand what information is learned from this representation. Figure 12 summarizes this investigation. Plot a shows the layout with points colored by their residual value - darker color indicates larger residual and poor fit. The plots b, c, d, e show samples of hand-written digits taken from inside the colored boxes. Going from top to bottom around the curve shape, we can see that the “1”s are drawn ~~with~~ from right slant to a left slant. The “1”s in d (black box) tend to have the extra up stroke, but are quite varied in appearance. The “1”s shown in the plots labelled e correspond to points with big residuals. They can be seen to be more strangely drawn than the others. Overall, this 2- D layout shows a useful way to summarize the variation in way “1”s are drawn.

~~Summary from exploring tSNE layout of the MNIST digit 1 data (a in Figure 12) using linked brushing. There is a big nonlinear cluster (grey) and a small cluster (orange) located very close to the one corner of the big cluster in (a). The MNIST digit 1 data has a nonlinear structure in 10- D . Two projections from a tour on 10- D reveal that the small orange cluster is actually a diffuse set of points wrapped within the grey cluster, which is C-shaped in the high dimensions.~~

~~Summary of the layout structure, and large errors, relative to the MNIST digit 1 shape: (a) layout colored by residual value, and at right (b-e) are images of samples of observations~~

taken at locations around the layout, showing similarity in how the 1's were drawn. Set (f) are images corresponding to large residuals in the big cluster (darker orange in plot a). Along the big cluster, the angle of digit 1 changes (b-d). The small cluster has larger residuals, and the images show that these tend to be European style with a flag at the top, and a base at the bottom. The set in (f) show various poorly written digits.

9 Discussion

We have developed an approach to help assess and compare NLDR layouts, ~~generated by different methods and hyper-parameter choice(s).~~ It depends on conceptualizing the 2- D layout as a model, allowing for the creation of a wireframe representation of the model that can be lifted into p - D . ~~The fit is assessed by viewing the model in the data space, computing residuals and HBE. Different layouts can be compared using the HBE, providing a quantitative metric to decide on the most suitable NLDR layout to represent the data.~~ goodness of fit of the layout is measured by a new statistic, HBE, which can be used to compare multiple layouts. Global and local preservation of structure is assessed by examining the HBE across a range of binwidths. ~~It~~ Viewing the fitted model in the high-dimensional data space can help to determine how well it captures structures in the data. This approach also provides a simple way to predict the values of new p - D observations in the 2- D , which could be useful for implementing uncertainty checks, such as using training and testing samples.

~~The new methodology is accompanied by an R package called~~ This methodology is made available in the R package `quollr`, so that it is readily usable and broadly accessible. The

package has methods to fit the model, compute diagnostics, and also visualize the results, with interactivity. We have primarily used the `langevitour` software (Harrison 2023) to view the model in the data space, but other tour software such as `tourr` (Wickham et al. 2011) and `detourr` (Hart & Wang 2022) could also be used.

Two examples illustrating usage are provided: the PBMC3k data, where the NLDR is summarizing clustering in p -D, and MNIST data illustrating how NLDR represents an intrinsically low-dimensional nonlinear manifold. We examined a typical published usage of UMAP with illustrated that a published UMAP layout of the PBMC3k dataset (Chen et al. 2024). As is typical of UMAP layout with default settings, the separation between clusters is grossly exaggerated. The layout even suggests, is sub-optimal. The chosen layout inaccurately exaggerated separation between clusters, even suggesting separation where there is none. Our approach provides a way to choose a reasonable layout and avoids the use of misleading layouts in the future. was none. In the hand-written digits (LeCun et al. 1998), we illustrate how our model fit statistics show that a flat disc layout is superior to the curved-shaped layouts, and how illustrating how an NLDR can condense some sparse structure, making it appear as a cluster. This also illustrated the use of model residuals to identify oddly written “1”s using the residuals of the fitted model digits.

The supplementary material contains examples of strange behavior of NLDR methods, including twisting, flattened “pancake” clusters in p -D, and severe effects of density differences.

This work can be applied with The HBE metric complements existing metrics for evaluating

NLDR layout, ~~such as~~ ARNX, RTA, SC, and RGS. It ~~provides an additional evaluation metric, and importantly allows any layout to be viewed in the data space. This latter aspect can help to disentangle conflicting suggestions by the different metrics.~~

~~Additional exploration of~~ suggests that more work on distance measures to summarize the fit could be ~~a valuable direction for future work. We have used~~ valuable. HBE uses Euclidean distance, but other measures, such as geodesic distances (Tenenbaum et al. 2000), may ~~better capture~~ help capture departures from curved or nonlinear ~~relationships~~ surfaces in the data ~~and are worth exploring.~~

~~This work has also revealed some interesting behaviors of NLDR methods, including twisting, flattened “pancake” clusters in , and severe effects of density differences. These are described in more detail in the supplementary materials.~~

~~Researchers usually use layouts, but if a~~ The approach presented here could also be extended for k - D ($k > 2$) ~~layout is provided the approach developed here could be extended.~~ ~~Potential approaches include~~) layouts. Using 3- D binning, k -means clustering with nearest neighbors edge connection, or even special implementations of convex hulls, could provide model representations.

10 Supplementary Materials

All the materials to reproduce the paper can be found at <https://github.com/JayaniLakshika/paper-nldr-vis-algorithm>.

The supplementary materials provide additional details on the methods and hyperparameters used to generate layouts, video links of animated p - D tours, notation summaries,

and the R and Python scripts used in the study. They also describe the generation of the 2NC7 data, computation of hexagon grid configurations, and data binning procedures. Further sections highlight interesting NLDR behaviors observed in the data space and compare HBE with existing evaluation metrics for the PBMC3k and MNIST datasets.

The R package `quollr`, available on CRAN and at <https://jayanilakshika.github.io/quollr/>, provides software accompanying this paper to fit the wireframe model representation, compute diagnostics, visualize the model in the data with `langevitour`, and link multiple plots interactively.

11 Acknowledgments

These R packages were used for the work: `tidyverse` (Wickham et al. 2019), `Rtsne` (Krijthe 2015), `umap` (Konopka 2023), `patchwork` (Pedersen 2024), `colorspace` (Zeileis et al. 2020), `langevitour` (Harrison 2023), `conflicted` (Wickham 2023), `reticulate` (Ushey et al. 2024), `kableExtra` (Zhu 2024). These python packages were used for the work: `trimap` (Amid & Warmuth 2022) and `pacmap` (Wang et al. 2021). The article was created with R packages `quarto` (Allaire & Dervieux 2024).

References

10x Genomics (2016), ‘3k PBMCs from a Healthy Donor, Universal 3’ Gene Expression dataset analysed using Cell Ranger 1.1.0’. Accessed: 2025-06-09. <https://www.10xgenomics.com/datasets/3-k-pbm-cs-from-a-healthy-donor-1-standard-1-1-0>.

- Allaire, J. & Dervieux, C. (2024), *quarto: R Interface to Quarto Markdown Publishing System*. R package version 1.4.4. <https://CRAN.R-project.org/package=quarto>.
- Amid, E. & Warmuth, M. K. (2022), ‘TriMap: Large-scale Dimensionality Reduction Using Triplets’, *arXiv preprint arXiv:1910.00204* .
- Asimov, D. (1985), ‘The Grand Tour: A Tool for Viewing Multidimensional Data’, *SIAM Journal of Scientific and Statistical Computing* **6**(1), 128–143.
- Borg, I. & Groenen, P. J. F. (2005), *Modern Multidimensional Scaling*, 2nd edn, Springer, New York, NY. <https://doi.org/10.1007/0-387-28981-X>.
- Brown, T. A. (2015), *Confirmatory Factor Analysis for Applied Research*, 2nd edn, The Guilford Press, New York, NY, US.
- Carr, D. B., Littlefield, R. J., Nicholson, W. L. & Littlefield, J. S. (1987), ‘Scatterplot Matrix Techniques for Large N’, *Journal of the American Statistical Association* **82**(398), 424–436. <http://www.jstor.org/stable/2289444>.
- Carr, D., Lewin-Koh, N., Maechler, M. & Sarkar, D. (2023), *hexbin: Hexagonal Binning Routines*. R package version 1.28.3. <https://CRAN.R-project.org/package=hexbin>.
- Chari, T. & Pachter, L. (2023), ‘The specious art of single-cell genomics’, *PLoS Computational Biology* **19**, e1011288. <https://doi.org/10.1371/journal.pcbi.1011288>
- Chen, Z., Wang, C., Huang, S., Shi, Y. & Xi, R. (2024), ‘Directly Selecting Cell-type Marker Genes for Single-cell Clustering Analyses’, *Cell Reports Methods* **4**(7), 100810. <https://www.sciencedirect.com/science/article/pii/S2667237524001735>.

- Cheng, J. & Sievert, C. (2025), *crosstalk: Inter-Widget Interactivity for HTML Widgets*. R package version 1.2.2. <https://CRAN.R-project.org/package=crosstalk>
- Cheng, J., Sievert, C., Schloerke, B., Chang, W., Xie, Y. & Allen, J. (2024), *htmltools: Tools for HTML*. R package version 0.5.8.1. <https://CRAN.R-project.org/package=htmltools>
- Coifman, R. R., Lafon, S., Lee, A. B., Maggioni, M., Nadler, B., Warner, F. & Zucker, S. W. (2005), ‘Geometric Diffusions as a Tool for Harmonic Analysis and Structure Definition of Data: Diffusion Maps’, *Proceedings of the National Academy of Sciences of the United States of America* **102**(21), 7426–7431. <https://www.pnas.org/doi/abs/10.1073/pnas.0500334102>.
- Freedman, D. A. & Diaconis, P. (1981), ‘On the Histogram as a Density Estimator: L2 Theory’, *Probability Theory and Related Fields* **57**, 453–476. <https://doi.org/10.1007/BF01025868>.
- Gebhardt, A., Bivand, R. & Sinclair, D. (2024), *interp: Interpolation Methods*. R package version 1.1-6. <https://CRAN.R-project.org/package=interp>.
- Haque, A., Engel, J., Teichmann, S. A. & Lönnberg, T. (2017), ‘A Practical Guide to Single-cell RNA-sequencing for Biomedical Research and Clinical Applications’, *Genome Medicine* **9**(1), 75. <https://doi.org/10.1186/s13073-017-0467-4>.
- Harrison, P. (2023), ‘langevitour: Smooth Interactive Touring of High Dimensions, Demonstrated With scRNA-Seq Data’, *The R Journal* **15**, 206–219. <https://doi.org/10.32614/RJ-2023-046>.
- Hart, C. & Wang, E. (2022), *detourr: Portable and Performant Tour Animations*. R package

version 0.1.0. <https://casperhart.github.io/detourr/>.

Hinton, G. E. & Salakhutdinov, R. R. (2006), ‘Reducing the dimensionality of data with neural networks’, *Science* **313**(5786), 504–507.

Irizarry, R. (2024), ‘Biologists, stop putting umap plots in your papers’, <https://simplystatistics.org/posts/2024-12-23-biologists-stop-including-umap-plots-in-your-papers/>. Blog post on Simply Statistics.

Johnstone, I. M. & Titterton, D. M. (2009), ‘Statistical Challenges of High-Dimensional Data’, *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* **367**(1906), 4237–4253. <https://royalsocietypublishing.org/doi/abs/10.1098/rsta.2009.0159>.

Jolliffe, I. (2011), *Principal Component Analysis*, Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 1094–1096. https://doi.org/10.1007/978-3-642-04898-2_455.

Jöreskog, K. G. (1969), ‘A General Approach to Confirmatory Maximum Likelihood Factor Analysis’, *Psychometrika* **34**(2), 183–202. <https://doi.org/10.1007/BF02289343>.

Koch, I. (2013), *Analysis of Multivariate and High-Dimensional Data*, Cambridge Series in Statistical and Probabilistic Mathematics, Cambridge University Press.

Konopka, T. (2023), *umap: Uniform Manifold Approximation and Projection*. R package version 0.2.10.0. <https://CRAN.R-project.org/package=umap>.

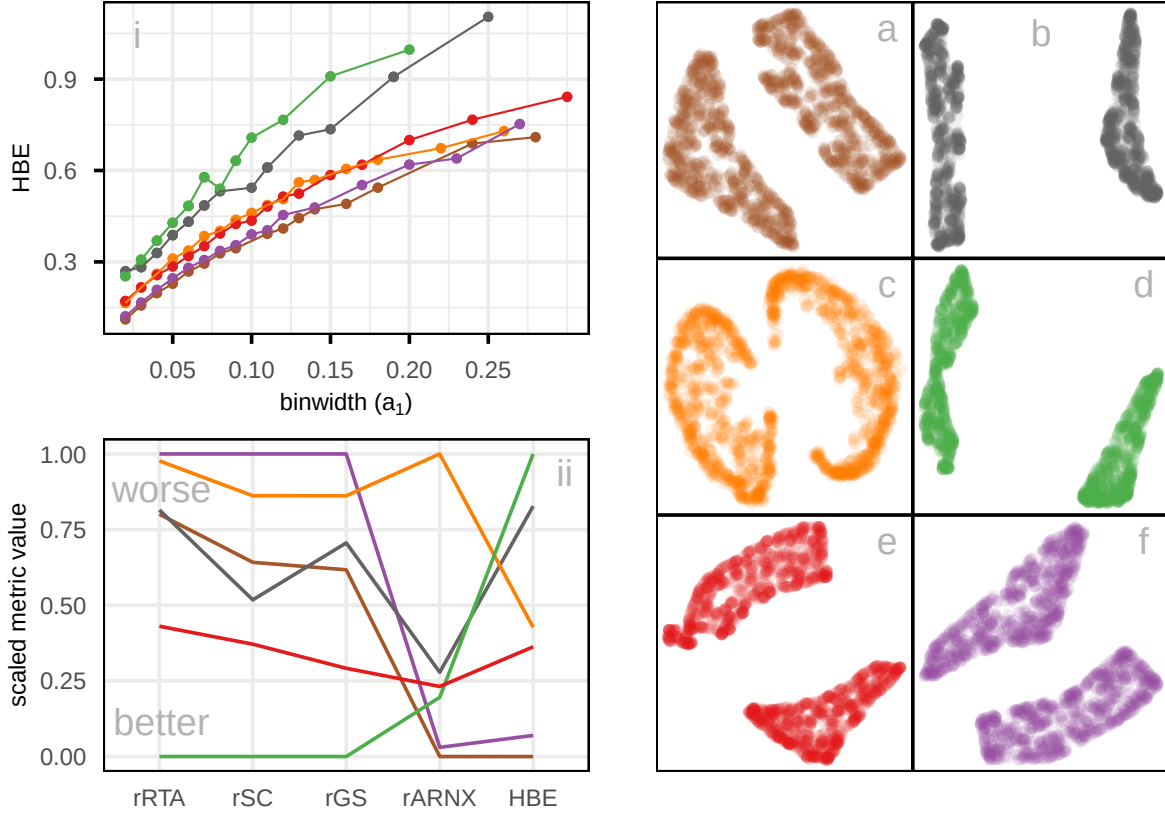
Krijthe, J. H. (2015), *Rtsne: T-Distributed Stochastic Neighbor Embedding Using Barnes-Hut Implementation*. R package version 0.16. <https://github.com/jkrijthe/Rtsne>.

- Kruskal, J. B. (1964), ‘Nonmetric Multidimensional Scaling: A Numerical Method’, *Psychometrika* **29**(2), 115–129. <https://doi.org/10.1007/BF02289694>.
- Laa, U., Cook, D. & Lee, S. (2022), ‘Burning Sage: Reversing the Curse of Dimensionality in the Visualization of High-Dimensional Data’, *Journal of Computational and Graphical Statistics* **31**(1), 40–49. <https://doi.org/10.1080/10618600.2021.1963264>.
- LeCun, Y., Cortes, C. & Burges, C. J. C. (1998), ‘The MNIST Database of Handwritten Digits’. Accessed: 2025-06-09. <http://yann.lecun.com/exdb/mnist/>.
- Lee, D. T. & Schachter, B. J. (1980), ‘Two Algorithms For Constructing a Delaunay Triangulation’, *International Journal of Computer & Information Sciences* **9**(3), 219–242. <https://doi.org/10.1007/BF00977785>.
- Lee, J. A., Peluffo-Ordóñez, D. H. & Verleysen, M. (2015), ‘Multi-scale similarities in stochastic neighbour embedding: Reducing dimensionality while preserving both local and global structure’, *Neurocomputing* **169**, 246–261. <https://www.sciencedirect.com/science/article/pii/S0925231215003641>
- Lee, S., Cook, D., da Silva, N., Laa, U., Wang, E., Spyrisson, N. & Zhang, H. S. (2021), ‘A Review of the State-of-the-Art on Tours for Dynamic Visualization of High-Dimensional Data’, *arXiv preprint arXiv:2104.08016*.
- Lee, S., Laa, U. & Cook, D. (2022), ‘Casting multiple shadows: interactive data visualisation with tours and embeddings’, *Journal of Data Science, Statistics, and Visualisation* **2**(3). <https://jdssv.org/index.php/jdssv/article/view/21>
- Maaten, L. V. D. & Hinton, G. E. (2008), ‘Visualizing Data Using t-SNE’, *Journal of Machine*

- Learning Research* **9**(11), 2579–2605. <https://api.semanticscholar.org/CorpusID:5855042>.
- McInnes, L., Healy, J., Saul, N. & Großberger, L. (2018), ‘UMAP: Uniform Manifold Approximation and Projection’, *Journal of Open Source Software* **3**(29), 861. <https://doi.org/10.21105/joss.00861>.
- Moon, K. R., van Dijk, D., Wang, Z., Gigante, S. A., Burkhardt, D. B., Chen, W. S., Yim, K., van den Elzen, A., Hirn, M. J., Coifman, R. R., Ivanova, N. B., Wolf, G. & Krishnaswamy, S. (2019), ‘Visualizing Structure and Transitions in High-Dimensional Biological Data’, *Nature Biotechnology* **37**, 1482–1492. <https://doi.org/10.1038/s41587-019-0336-3>.
- Pedersen, T. L. (2024), *patchwork: The Composer of Plots*. R package version 1.2.0. <https://CRAN.R-project.org/package=patchwork>.
- Saeed, N., Nam, H., Haq, M. I. U. & Muhammad Saqib, D. B. (2018), ‘A Survey on Multidimensional Scaling’, *ACM Comput. Surv.* **51**(3). <https://doi.org/10.1145/3178155>.
- Shepard, R. N. (1962), ‘The analysis of proximities: Multidimensional scaling with an unknown distance function. i.’, *Psychometrika* **27**(2), 125–140.
- Sievert, C. (2020), *Interactive Web-Based Data Visualization with R, plotly, and shiny*, Chapman and Hall/CRC. <https://plotly-r.com>
- Silva, V. & Tenenbaum, J. (2002), ‘Global Versus Local Methods in Non-linear Dimensionality Reduction’, *Advances in Neural Information Processing Systems* **15**, 721–728. https://proceedings.neurips.cc/paper_files/paper/2002/file/5d6646aad9bcc0be55b2c82f69750387-Paper.pdf.
- Spearman, C. (1961), ‘The proof and measurement of association between two things.’

- Tenenbaum, J. B., de Silva, V. & Langford, J. C. (2000), ‘A global geometric framework for nonlinear dimensionality reduction’, *Science* **290**(5500), 2319–2323. <https://www.science.org/doi/abs/10.1126/science.290.5500.2319>
- Ushey, K., Allaire, J. & Tang, Y. (2024), *reticulate: Interface to Python*. R package version 1.38.0. <https://CRAN.R-project.org/package=reticulate>.
- van der Maaten, L. (2009), Learning a parametric embedding by preserving local structure, in ‘Artificial Intelligence and Statistics’, PMLR, pp. 384–391.
- Wang, Y., Huang, H., Rudin, C. & Shaposhnik, Y. (2021), ‘Understanding How Dimension Reduction Tools Work: An Empirical Approach to Deciphering t-SNE, UMAP, TriMap, and PaCMAP for Data Visualization’, *Journal of Machine Learning Research* **22**(201), 1–73. <http://jmlr.org/papers/v22/20-1061.html>.
- Wickham, H. (2023), *conflicted: An Alternative Conflict Resolution Strategy*. R package version 1.2.0. <https://CRAN.R-project.org/package=conflicted>.
- Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Golemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K. & Yutani, H. (2019), ‘Welcome to the Tidyverse’, *Journal of Open Source Software* **4**(43), 1686. <https://doi.org/10.21105/joss.01686>.
- Wickham, H., Cook, D. & Hofmann, H. (2015), ‘Visualizing Statistical Models: Removing the Blindfold’, *Statistical Analysis and Data Mining: The ASA Data Science Journal* **8**(4), 203–225. <https://onlinelibrary.wiley.com/doi/abs/10.1002/sam.11271>.

- Wickham, H., Cook, D., Hofmann, H. & Buja, A. (2011), ‘tourr: An R Package For Exploring Multivariate Data With Projections’, *Journal of Statistical Software* **40**(2), 1–18. <http://www.jstatsoft.org/v40/i02/>.
- Zeileis, A., Fisher, J. C., Hornik, K., Ihaka, R., McWhite, C. D., Murrell, P., Stauffer, R. & Wilke, C. O. (2020), ‘colorspace: A Toolbox for Manipulating and Assessing Colors and Palettes’, *Journal of Statistical Software* **96**(1), 1–49. <https://www.jstatsoft.org/index.php/jss/article/view/v096i01>.
- Zhu, H. (2024), *kableExtra: Construct Complex Table with kable and Pipe Syntax*. R package version 1.4.0. <https://CRAN.R-project.org/package=kableExtra>.



Assessing which of the 6 NLDR layouts (a-f) on the 2NC7 data is the better representation using HBE for varying (i) binwidth (a_1), and (ii) average bin count ($\bar{n}_h$). Color represents NLDR layout. Layout d is universally poor. Layouts b, e that show two close clusters are universally suboptimal. Layout f with little separation performs well at tiny binwidth (where most points are in their own bin) and poorly as binwidth increases. Layout e has small separation with oddly shaped clusters. Layout a is the best choice. Plot (ii), which compares HBE values with respect to average bin count, helps account for differences in cluster density; here, the variation among layouts is reduced, showing that some differences observed in (i) arise from density rather than true structure. Comparison of scaled evaluation metrics (rRTA, rSC, rGS, rARNX, and HBE using $a_1 = 0.05$) for the six NLDR layouts computed on the 2NC7 data using a parallel coordinate plot (iii). Color of the line indicates NLDR layout.

Figure 6: Assessing which of the 6 NLDR layouts (a-f) on the 2NC7 data is the better representation using (i) HBE for varying binwidth (a_1), and (ii) scaled evaluation metrics (rRTA, rSC, rGS, rARNX, and HBE using $a_1 = 0.05$). Color represents the NLDR layout. Using HBE (i), layouts a and f are nearly indistinguishable from each other but have universally lower values than the others: either layout could adequately represent the data. HBE mostly agrees with rARNX, except for layout d. The other metrics rank the layouts in reverse order! The problem, though, is that they favor more separation, which is misleading for this data. This can be validated by viewing the data in a tour.

When the metrics disagree it causes confusion for the analyst, and thus provides a temptation to choose the nicest looking layout (very separated clusters), even though it may be a hallucination. Because HBE is accompanied with a representation of the

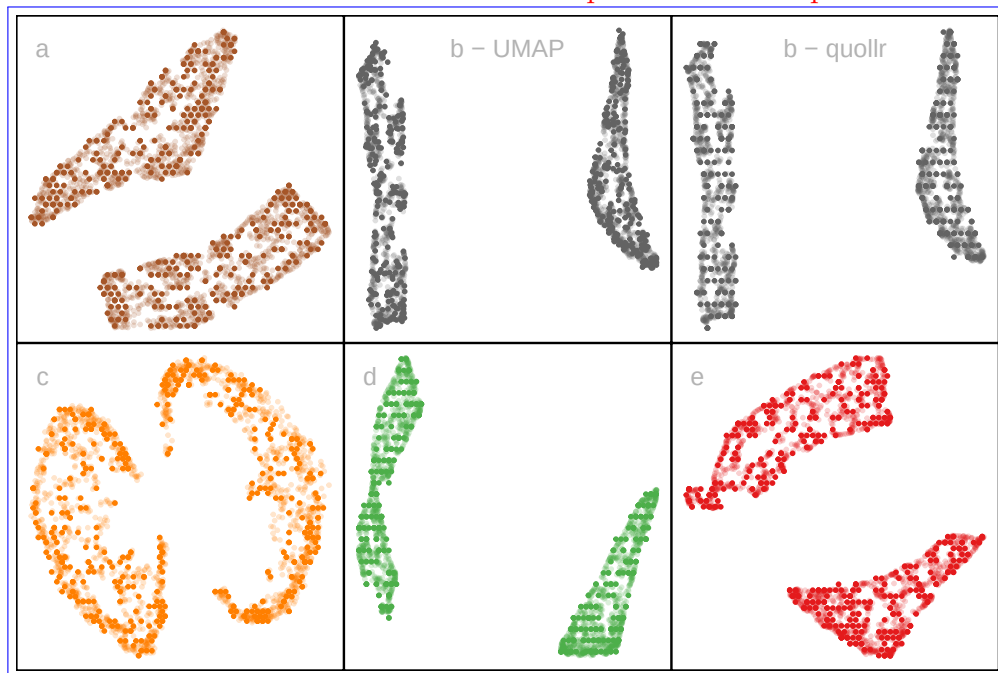


Figure 7: Predicted test data (dark) overlaid on training sets (light), for (most) layouts shown in Figure 6. UMAP is the only method to provide predicted test set locations, and these are compared with our approach in panels (b - UMAP) and (b - quollr). All other panels show predictions obtained using our method applied. The predicted test set data match the training set layouts very neatly.

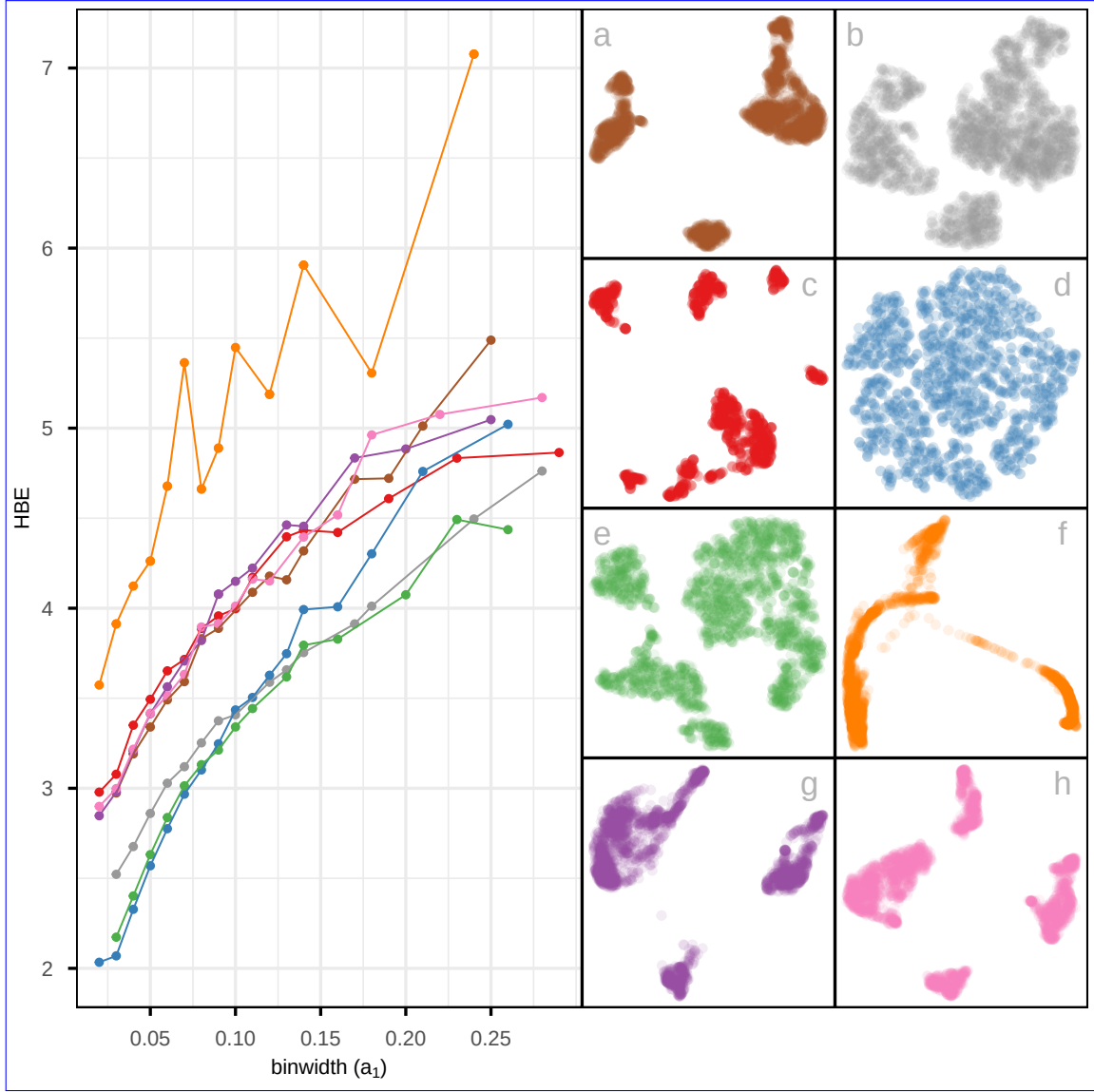


Figure 8: Assessing which of the 8 NLDR layouts on the PBMC3k data (shown in Figure 1) is the better representation using HBE for varying (i) binwidth (a_1), and (ii) average bin count ($\bar{n}_h$). Color used for the lines and points in the left plot and in the scatterplots represents the NLDR layout (a-h). Layout f is universally poor. Layouts a, c, g, and h that show large separations between clusters are universally suboptimal. Layout d with little separation performs well at tiny binwidth (where most points are in their own bin) and poorly as binwidth increases. The choice of the best is between layouts b and e, which have small separations between oddly shaped clusters. Layout e is chosen as the best. Plot (ii), which accounts for the density within clusters by using average bin count, shows reduced differences between layouts, indicating that part of the variation in (i) is driven by cluster density rather than true structural differences.

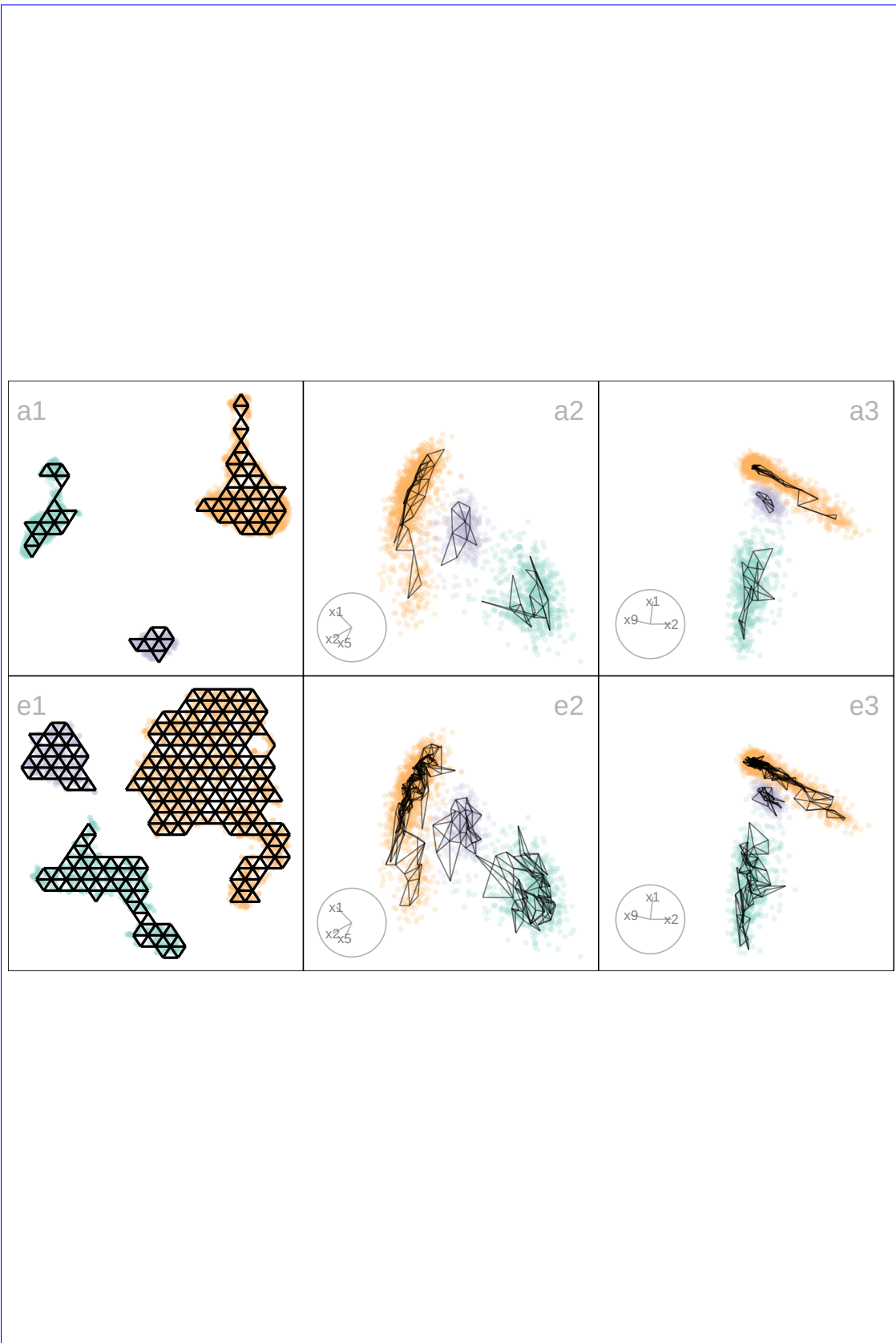


Figure 9: Compare the published 2-D layout (a) made with UMAP and the 2-D layout selected by HBE plot (e) made by tSNE. The two plots on the right show projections from a tour, with the models overlaid. The published layout suggests three very separated clusters, but this is not present in the data. While there may be three clusters, they are not well-separated. The difference in model fit also indicates this: the published layout does not spread out fully into the point cloud like the model generated from layout e. This supports the choice that layout e is the better representation of the data, because it does

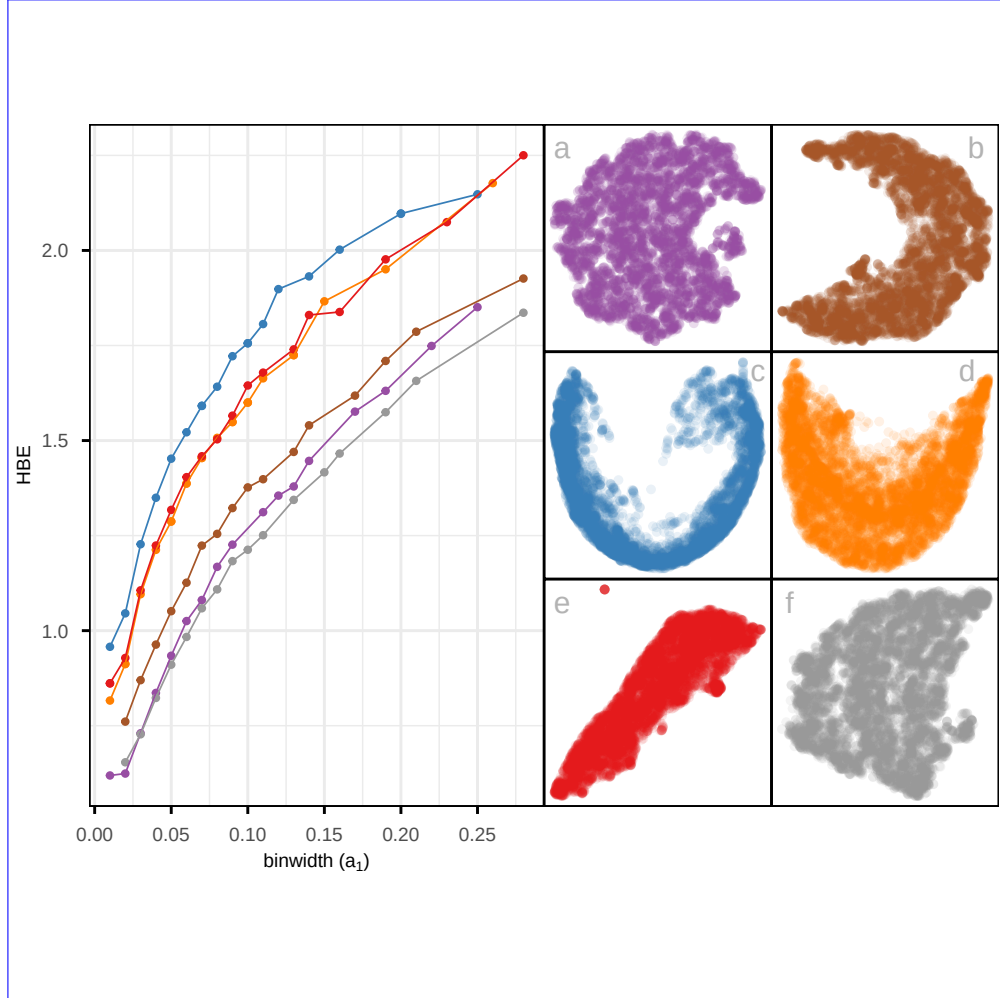


Figure 10: Assessing which of the 6 NLDR layouts of the MNIST digit 1 data is the better representation using HBE for varying (i) binwidth (a_1), and (ii) average bin count ($\bar{n}_h$). ~~Colour~~-Color is used for the lines and points in the left plot to match the scatterplots of the NLDR layouts (a-f). Layout c is universally poor. Layouts a, b, and f that show a big cluster and a small circular cluster are universally optimal. Layout a performs well at tiny binwidth (where most points are in their own bin) and not as well as f with larger binwidth, thus layout f is the best choice. Plot (ii), which accounts for the density within clusters by using average bin count, shows reduced differences between layouts, indicating that part of the variation in (i) is driven by cluster density rather than true structural differences.

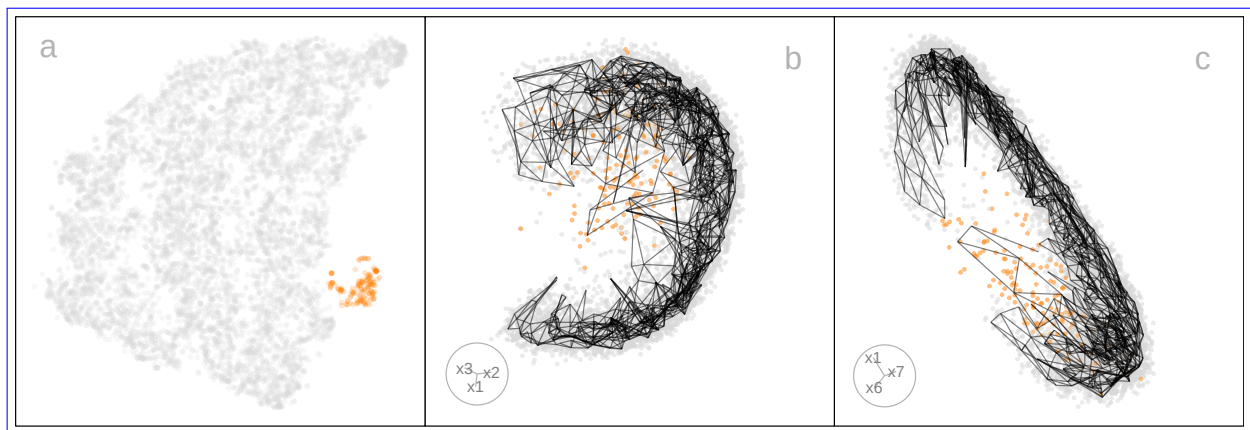


Figure 11: Summary from exploring tSNE layout of the MNIST digit 1 data (a in Figure 12) using linked brushing. There is a big nonlinear cluster (grey) and a small cluster (orange) located very close to one corner of the big cluster in 2- D (a). The MNIST digit 1 data has a nonlinear structure in 10- D . Two 2- D projections from a tour on 10- D reveal that the small orange cluster is actually a diffuse set of points wrapped within the grey cluster, which is C-shaped in the high dimensions.

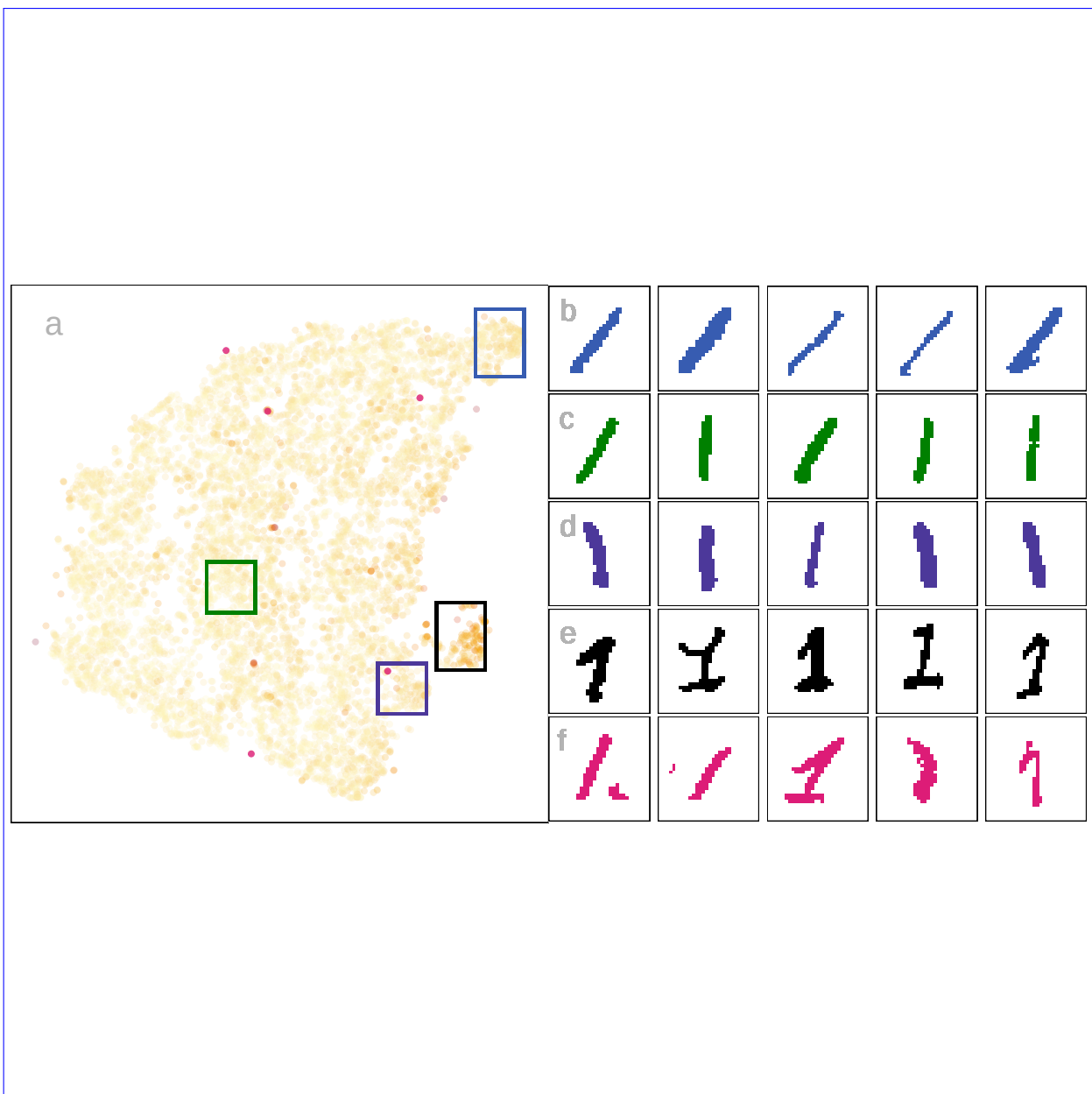


Figure 12: Summary of the layout structure, and large errors, relative to the MNIST digit 1 shape: (a) layout colored by residual value, and at right (b-e) are images of samples of observations taken at locations around the layout, showing similarity in how the 1's were drawn. Set (f) are images corresponding to large residuals in the big cluster (darker pink in plot a). Along the big cluster, the angle of digit 1 changes (b-d). The small cluster has larger residuals, and the images show that these tend to be European style with a flag at the top and a base at the bottom. The set in (f) shows various poorly written digits.